

ASSIGNMENT (Aug 18, 2026)

RNA-Seq Literature Familiarization and Data Characterization

Preparation for RNA-Seq Re-analysis Using Galaxy and GitHub

Part 1. Form Four New Groups 1. Form four new groups

Members	Batigulao, Jehiah Bless Obiñeta, Inah Marie A. Oficiar, Francis Kyle Pasculado, Mark Ryan Suan, Jade Angela
Group Topic	Salinity Stress

Part 2. Find One Published RNA-Seq Study

Transcriptomic analysis reveals candidate genes associated with salinity stress tolerance during the early vegetative stage in fababean genotype, Hassawi-2

[Muhammad Afzal](#)¹, [Salem S Alghamdi](#)¹, [Muhammad Altaf Khan](#)^{1,✉}, [Suliman A Al-Faifi](#)¹, [Muhammad Habib ur Rahman](#)^{2,3,✉}

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PMCID: PMC10692206 PMID: [38040745](#)

Abstract

Abiotic stresses are a significant constraint to plant production globally. Identifying stress-related genes can aid in the development of stress-tolerant elite genotypes and facilitate trait and crop manipulation. The primary aim of this study was to conduct whole transcriptome analyses of the salt-tolerant faba bean genotype, Hassawi-2, under different durations of salt stress (6 h, 12 h, 24 h, 48 h, and 72 h) at the early vegetative stage, to better understand the molecular basis of salt tolerance. After de novo assembly, a total of 140,308 unigenes were obtained. The up-regulated differentially expressed genes (DEGs) were 2380, 2863, 3057, 3484, and 4820 at 6 h, 12 h, 24 h, 48 h, and 72 h of salt stress, respectively. Meanwhile, 1974, 3436, 2371, 3502, and 5958 genes were downregulated at 6 h, 12 h, 24 h, 48 h, and 72 h of salt stress, respectively. These DEGs encoded various regulatory and functional proteins, including kinases, plant hormone proteins, transcriptional factors (TFs) basic helix-loop-helix (*bHLH*), *Myeloblastosis* (*MYB*), and (*WRKY*), *heat shock proteins* (*HSPs*), *late embryogenesis abundant* (*LEA*) proteins, dehydrin, antioxidant enzymes, and aquaporin

Figure 1. Published RNA-seq Study Article used in the activity entitled “Transcriptomic analysis reveals candidate genes associated with salinity stress tolerance during the early vegetative stage in faba bean genotype, Hassawi-2”.

Part 3. Understand the Paper Before Downloading Data

Information from the Paper	Group Answer
Title of the paper	Transcriptomic analysis reveals candidate genes associated with salinity stress tolerance during the early vegetative stage in fababean genotype, Hassawi-2
Authors	Muhammad Afzal, Salem S. Alghamdi, Muhammad Altaf Khan, Sulieman A. Al-Faifi, & Muhammad Habib ur Rahman
Year	2023
Journal	<i>Scientific Reports</i> , 13, 21223
Organism	Faba bean (<i>Vicia faba</i> L.), salt-tolerant genotype Hassawi-2
Tissue / organ / cell type	Leaves at the early vegetative/three-leaf stage
Biological question	To identify genes and molecular mechanisms involved in salinity stress tolerance in Hassawi-2 faba bean at different durations of salt exposure.
Control condition	Faba bean plants grown without the 200 mM NaCl salt treatment
Treatment or stress condition	200 mM NaCl salt stress for 6, 12, 24, 48, and 72 hours
Number of biological replicates	3 biological replicates per condition/time point
Sequencing platform	Illumina NovaSeq 6000 (sequencing outsourced to Macrogen)
Single-end or paired-end sequencing	Paired-end
Read length, if reported	101 bp per read
RNA-seq repository	NCBI Sequence Read Archive (SRA)
BioProject / Study accession	PRJNA943415 (SRA submission ID SUB12947558)
Run accession numbers	Not listed individually in the paper — retrieve the SRR list from the BioProject/SRA page for PRJNA943415
Reference genome or transcriptome used by the authors	None, de novo assembly (no reference genome used; faba bean genome had only just been published)
Main RNA-seq software or tools reported by the authors	Trimmomatic (adapter/quality trimming) → Trinity (de novo assembly) → CD-HIT-EST (clustering into unigenes) → TransDecoder (ORF prediction) → Bowtie (read alignment)

	→ RSEM (expression/FPKM quantification) → edgeR (DEG analysis, TMM normalization) → BLASTX/DIAMOND against NR, Swiss-Prot, KEGG, GO, COG/KOG for annotation
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Part 4. Prepare a Simple RNA-Seq Experimental Design Diagram

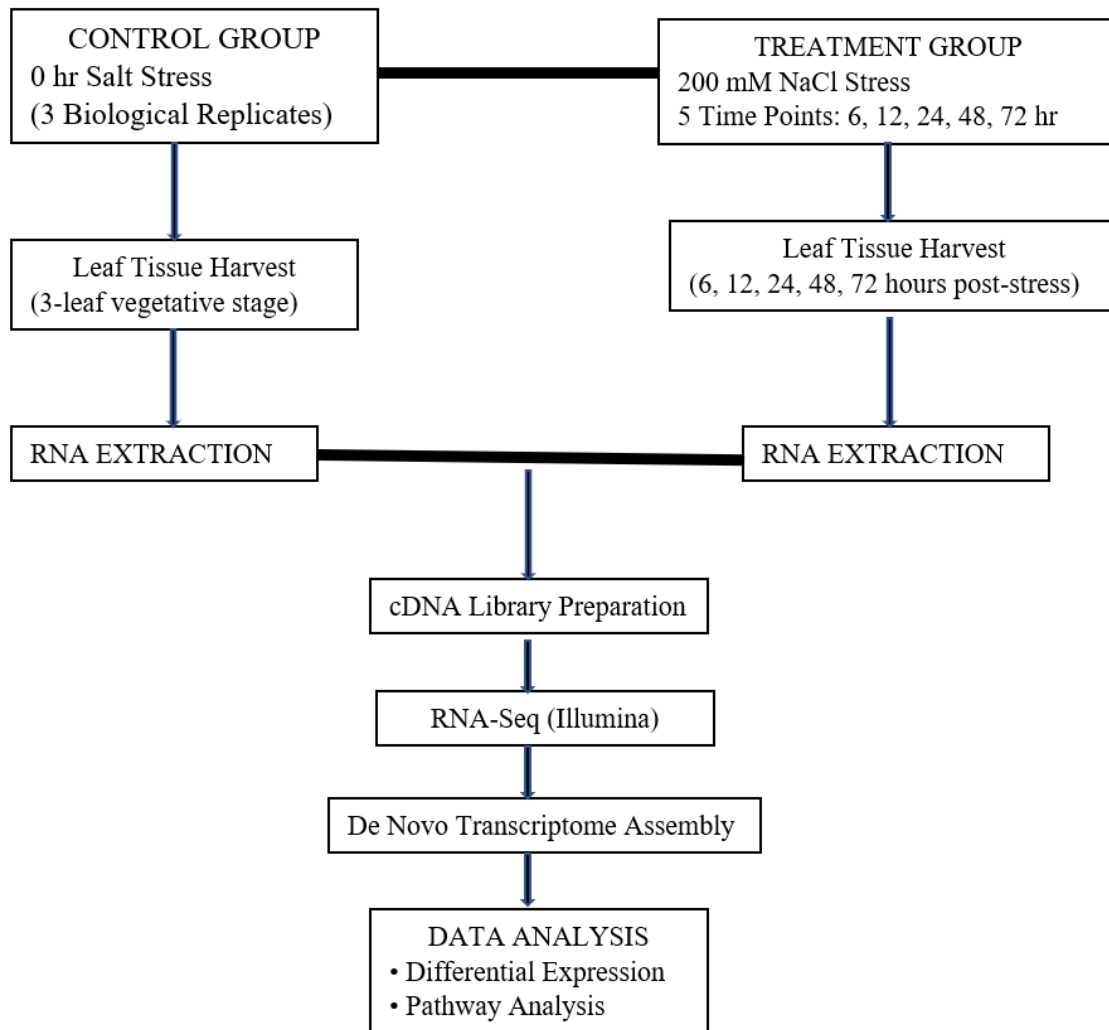


Figure 2. *Experimental Design of the study.*

- What was changed or manipulated?

The duration of exposure to salinity stress was the primary variable. Plants were exposed to 200 mM NaCl for five different time intervals to salinity stress was the primary variable. Plants were exposed to 200 mM NaCl for five different time intervals: 6 hours, 12 hours, 24 hours, 48 hours, and 72 hours.

- What was the control?

The control group consisted of faba bean plants at the same three-leaf vegetative stage grown under normal conditions (0 hours of salt stress).

- What was measured?

The whole transcriptome profile (gene expression levels) was measured using RNA sequencing. Specifically, the study identified 140,308 unigenes and quantified the Differentially Expressed Genes (DEGs) across all time points.

- How many biological replicates were used?

The study utilized three biological replicates for each sample group. A total of 17 libraries were sequenced (indicating that while three replicates were planned for all six conditions, one replicate was likely excluded during quality control).

- What tissue or cell type was sequenced?

Leaf tissue collected during the early vegetative stage (specifically at the three-leaf stage) was sequenced.

Part 5. Locate the Public RNA-Seq Data

#	Condition	Rep	SRX	Spots	Bases	Pair length (bases/spots)	Per-mate read length
1	Control	R1	SRX19681425	28.2M	5.7G	202.1 bp	101 bp
2	Control	R2	SRX19681431	27.9M	5.6G	200.7 bp	100 bp
3	Control	R3	SRX19681439	28.9M	5.8G	200.7 bp	100 bp
4	Stress 6h	R1	SRX19681440	28.2M	5.7G	202.1 bp	101 bp
5	Stress 6h	R2	SRX19681441	27.7M	5.6G	202.2 bp	101 bp
6	Stress 6h	R3	SRX19681426	25.4M	5.1G	200.8 bp	100 bp
7	Stress 12h	R1	SRX19681427	24.6M	5.0G	203.3 bp	102 bp
8	Stress 12h	R2	SRX19681428	22.4M	4.5G	200.9 bp	100 bp
9	Stress 12h	R3	SRX19681429	28.3M	5.7G	201.4 bp	101 bp
10	Stress 24h	R1	SRX19681430	31.6M	6.4G	202.5 bp	101 bp
11	Stress 24h	R2	SRX19681432	31.5M	6.4G	203.2 bp	102 bp
12	Stress 24h	R3	SRX19681433	26.3M	5.3G	201.5 bp	101 bp
13	Stress 48h	R1	SRX19681434	31.5M	6.4G	203.2 bp	102 bp
14	Stress 48h	R3	SRX19681436	22.5M	4.5G	200.0 bp	100 bp
15	Stress 72h	R1	SRX19681435	31.5M	6.4G	203.2 bp	102 bp
16	Stress 72h	R2	SRX19681437	31.3M	6.3G	201.3 bp	101 bp
17	Stress 72h	R3	SRX19681438	29.5M	6.0G	203.4 bp	102 bp

Part 6. Assign One RNA-Seq Sample to Each

Student	Run accession	Condition	Bio. replicate	Single/ Paired-end	Approx. file size
Obiñeta	SRX19681425 (SRR23869388)	Control	Rep 1	Paired-end	1.6 Gb
Suan	SRX19681440 (SRR23869373)	Treatment — 6h salt stress	Rep 1	Paired-end	1.6 Gb
Batigulao	SRX19681430 (SRR23869383)	Treatment — 24h salt stress	Rep 1	Paired-end	1.8Gb
Pasculado	SRX19681434 (SRR23869379)	Treatment — 48h salt stress	Rep 1	Paired-end	1.8 Gb
Oficiar	SRX19681435 (SRR23869378)	Treatment — 72h salt stress	Rep 1	Paired-end	1.8 Gb

Part 7. Download the RNA-Seq Data into Galaxy

Job Information ok

Galaxy Tool ID

toolshed.g2.bx.psu.edu/repos/iuc/sra_tools/fastq_dump/3.1.1+galaxy1

Job State

ok

Created

Wednesday Aug 26th 8:25:39 2026 GMT+8

Updated

Wednesday Aug 26th 10:03:00 2026 GMT+8

Time To Finish

1 hour 37 minutes 21 seconds

Command Line

mkdir -p ~/.ncbi && cp ~/expanselustre/scratch/sgalaxy/temp_project/jobs/main/79366100/...

ToolStandard.Output

Downloading accession: SRR23869383... 2026-08-26T01:11:26 prefetch.3.1.1: 1) Resolving '...

ToolStandard.Error

empty

Tool.Exit.Code

0

Job API ID

bbd44e69cb8906b56ddc12ba7f04db2b

Job Parameters

Input Parameter	Value
select input type	accession_number
Accession	SRR23869383
Select output format	gzip compressed fastq
adv	
Define format specification for sequence	@\$.Ssn[.Sm]/\$.ri
Minimum spot ID	Not available.
Maximum spot ID	Not available.
Minimum read length	Not available.
Split spot by read pairs	true
Output aligned or unaligned reads	both
aligned region	Not available.

History

search datasets

Assignment3_RNASeq_BATIGULAO_Group4

Default Storage

2.64 GB

8 4

10: FastQC on dataset 7: RawData

9: FastQC on dataset 7: Web page

6: collection 1 (reverse)

a list with 1 fastqsanger.gz dataset

5: collection 1 (forward)

a list with 1 fastqsanger.gz dataset

2: Single-end data (fastq-dump)

a list with 0 datasets

1: Paired-end data (fastq-dump)

a list with 1 fastqsanger.gz pair

Figure 3. Successful download and extract of Paired-end and Single-End RNA Seq Reads in Galaxy using the tool download and extract reads in FASTQ format from NCBI SRA.

7: SRR23869383:forward

ok

1.3 GB
format **fastqsanger.gz** database ? size 1.3 GB

Preview Visualize Details Edit

This dataset is large and only the first 100 KB is shown below.
Download

```

@SRR23869383.1/1
NCGGATTCGACTTAGAGCGTTCAGTCATAATCCAACGCACGGTAGCTTCGCCACTGGCTTTCAACCAAGCGCGATG
+
#FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@SRR23869383.2/1
NTTCCTCTTGTTTTCTCTCATCCATCAAATGTTCAAATCCTTCTCAGATCCGAAATTCGTTGAAGCAGAAC
+
#FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@SRR23869383.3/1
NCCGATGGTAAGTATCTTAGGAGAAGAAACAAGCAAAAAAATGCAGTTGTGCAATATTCGCCAAAAAATGCATAAA
+
#FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@SRR23869383.4/1
NTCCATTGAATGCATGTATATGCTGTAGCATATGAAATCCTTAAAACTCATCTGTATGAATCAGGCGCTTGGTTTG
+
#FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@SRR23869383.5/1
NGGTTCTCTGCCCTACGCGGAGGCTCTGGGGAAGTCAGAATAGGAGAGCACTCATCTGGGGTGGGCTTACTACTTAGAI
+
#FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@SRR23869383.6/1
NCTGGCTCCAGTGTCAACTGGCCTTCCGAATTCTGAGTCTCCAAATGAGTGTGTAATGCAITTTGTTCCGTG
+
#FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF

```

History

search datasets

Assignment3_RNASeq_BATIGULAO_Group4

Default Storage

2.64 GB

bpae

10: FastQC on dataset 7: RawData

9: FastQC on dataset 7: Web page

6: collection 1 (reverse)

5: collection 1 (forward)

2: Single-end data (fastq-du mp)

1: Paired-end data (fastq-du mp)

Figure 4. Preview of the forward dataset (*Treatment_24hrs_Rep1_R1*) from paired-end RNA sequencing data in FASTQ format using the tool Unzip collection into forward and reverse datasets.

8: SRR23869383:reverse

ok

1.3 GB
format **fastqsanger.gz** database ? size 1.3 GB

Preview Visualize Details Edit

This dataset is large and only the first 100 KB is shown below.
Download

```

@SRR23869383.1/2
GATGACAGTGTGCGAATAGTAATCAACCTAGTACGAGAGGAACCGTTGATTGCGACAATTGGTCATCGCGCTTGGTTGAA
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@SRR23869383.2/2
GTAGCTTTGATGATGGCTTGTTCGGGTGGAATCTTCTGCTTCAACGAAATTCGGGAGTCTGGATCTGAGAAGGATTI
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@SRR23869383.3/2
ACCCCTCATCTTCATACAAGAGTGTATCTCTCCCAATTTGTACTCGGATGCAGATGATGAATTGGATTTCATGCTTTATC
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@SRR23869383.4/2
CAAAAATCTTGATCTCGATCACTGGTTGATGATGTCTCAAAAAGGTTGATGGTTAAAGCCAATAGCAGATGAGCTGGG
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@SRR23869383.5/2
CAGTTATCGTGCCAACGTAACGCTGGTAGCAAGTGCAGGAGCGGATAACTGCTGAAAGCATCTAAGTAGTAAGCCCAI
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@SRR23869383.6/2
CGATGAGGCTCAATTAGTACTGGTGCAAAAGTTACCAATGGAGGTTACACTACCCACCGTAACAACGGAACCAAAATG
+

```

History: Assignment3_RNASeq_BATIGULAO...

collection 1 (reverse)

a list with 1 fastqsanger.gz dataset

Download

1: SRR23869383

Figure 5. Preview of the reverse dataset (*Treatment_24hrs_Rep1_R2*) from paired-end RNA sequencing data in FASTQ format using the tool Unzip collection into forward and reverse datasets.

Part 8. Examine the FASTQ File

Matrix	TREATMENT24HRS_Rep1_R1	TREATMENT24HRS_Rep1_R2
Line 1: sequence identifier	@SRR23869383.1/1	@SRR23869383.1/2
Line 2: nucleotide sequence	NCGGATTCTGACTTAGAGGCG TTCAGTCATAATCCAACGCAC GGTAGCTTCGCGCCACTGGCT TTTCAACCAAGCGCGATGACC AATTGTGCGAATCAACG	GATGACAGTGTGCGCAATAGTA ATTCAACCTAGTACGAGAGGA ACCGTTGATTGCGCACAATTGG TCATCGCGCTTGTTGAAAAG CCAGTGGCGCGAAGCTA
Line 3: separator line	+	+
Line 4: quality-score characters	#FFFFFFFFFFFFFFFFFFFFF::F:F F:FFFFFFFFFFFFFFFFFFFFF:F,FFF FFFF:FFF:FFFFFFFFF,;FFFF:FF:F F:F:FFFFFFFFF:FFFFF	FFFFFFFFFFFFFFFFF,;FFFFFFFFF FFFFFFFFFFFFFFFFF:FFFFFFFFF FFFF:FFFFFFFFF:FFFFFFFFF FFFFFFFFFFFFFFFFF:FFFFF

Answer:

1. How is a FASTQ file different from the FASTA genome file used in the previous assignments?

FASTQ vs FASTA: FASTQ contains the nucleotide sequences and their quality scores, while FASTA contains only the nucleotide sequences.

2. What information is present in FASTQ that is not present in FASTA?

Information unique to FASTQ: FASTQ includes a quality score for each nucleotide, showing the reliability of the sequencing data.

3. Why are quality scores important in RNA-seq?

Importance of quality scores: They help identify low-quality reads or bases, allowing them to be filtered or trimmed to improve the accuracy of downstream RNA-seq analysis.

Part 9. Run FastQC



Figure 6. Webpage Data Output of the FastQC Analysis for the Forward Read (Treatment_24hrs_Rep1_R1) Generated Using the FastQC Tool in Galaxy.

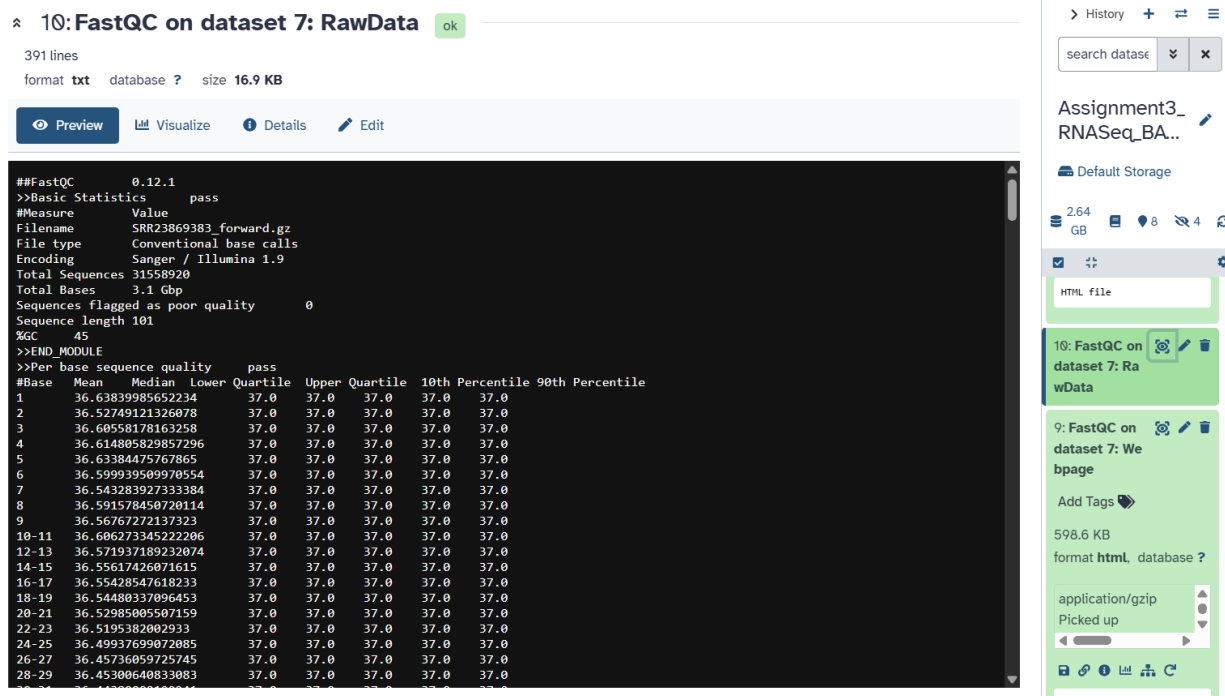


Figure 7. Raw Data Output of the FastQC Analysis for the Forward Read (Treatment_24hrs_Rep1_R1) Generated Using the FastQC Tool in Galaxy.



Figure 8. Webpage Data Output of the FastQC Analysis for the Reverse Read (Treatment_24hrs_Rep1_R2) Generated Using the FastQC Tool in Galaxy.

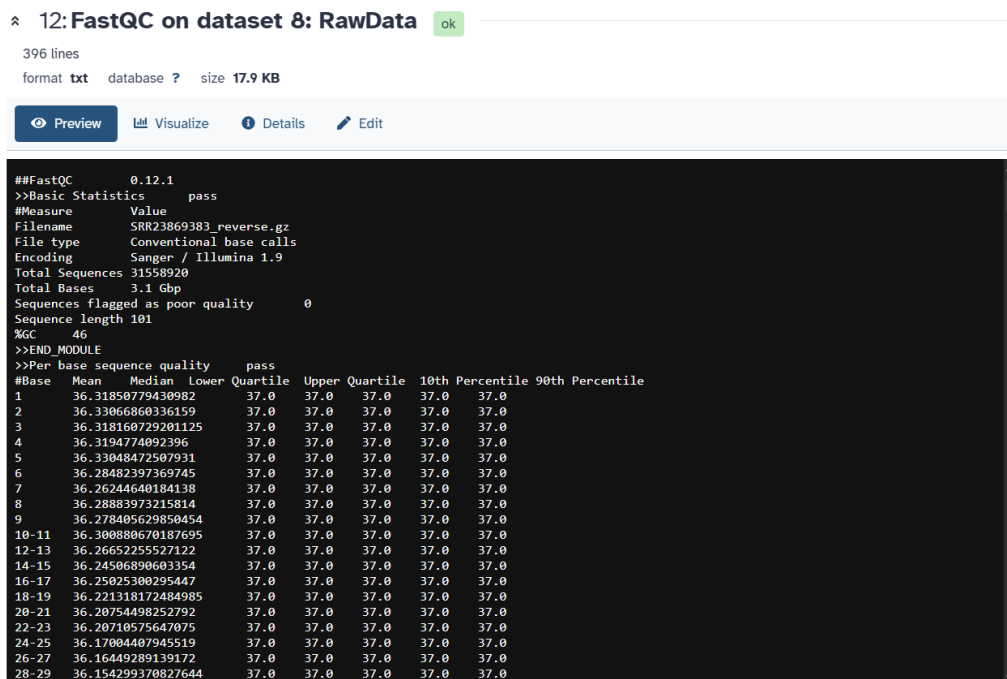


Figure 9. Raw Data Output of the FastQC Analysis for the Reverse Read (Treatment_24hrs_Rep1_R2) Generated Using the FastQC Tool in Galaxy.

Part 10. Characterize Your RNA-Seq Dataset

RNA-Seq Characteristic	Your Observation
RNA-seq run accession	R1&R2: SRR23869383
Condition	R1&R2: Treatment 24 hours
Single-end or paired-end	R1&R2: Paired-end
Number of reads / sequences	R1&R2: 31558920
Read length	R1&R2: 101bp
File size	R1&R2: 1.8Gb
GC content	R1: 45% & R2: 46%
General per-base sequence quality	R1&R2: pass
Presence of adapter content	R1&R2: pass
Presence of overrepresented sequences	R1: pass & R2: Warning — poly-G sequence detected (0.48%)
Any quality warning observed	Yes, R1 showed failures for per-base sequence content and sequence duplication levels. R2 showed warnings for per-base sequence content, per-sequence GC content, and overrepresented sequences, and failed sequence duplication levels.

Part 12. Compare the RNA-Seq Samples Within the Group

Student	Accession	Condition	Reads	Read length	GC %	Main QC observation
Obiñeta	SRR23869388	Control — 0h salt stress	28,209,473	101 bp	45%	Good quality, adapter-trimmed, uniform base composition; no major bias
Suan	SRR23869373	Treatment — 6h salt stress	28,211,353	101 bp	43-44%	Comparable to control; R2 has minor poly-G artifact (0.24%); all critical metrics pass
Batigulao	SRR23869383	Treatment — 24h salt stress	31,558,920	101 bp	45-46%	Main QC Observation: Approximately 45–46% of the reads

						passed the quality control, indicating moderate read quality. Further trimming and filtering may be needed before downstream analysis
Pasculado	SRR23869379	Treatment — 48h salt stress	31,549,121	101 bp	43-44%	Good overall quality with 31.5M reads and 43–44% GC, but FastQC showed per-base sequence content failure and a GC-content warning in R2
Oficiar	SRR23869378	Treatment — 72h salt stress	31,508,305	101 bp	43%	High duplication and sequence bias typical for RNA-seq, with significant Poly-G stretches in reverse reads caused by Illumina two-color chemistry.

Part 13. Interpretation Questions

1. What biological question was the original RNA-seq study trying to answer?

The study aimed to find out which genes and biological pathways help faba bean (genotype Hassawi-2) tolerate salt stress. By comparing gene expression in normal conditions versus salt stress over time (6 to 72 hours), the researchers wanted to understand how the plant responds to salinity at the molecular level and identify the specific genes that allow it to grow under high-salt conditions.

2. Why did the authors use RNA-seq instead of only examining the genome?

The genome contains all genes, but it does not show which genes are actually active or turned on at a given time or condition. RNA-seq measures gene expression, it tells us which genes are being transcribed into mRNA and at what levels. This is important because salinity stress changes how genes are expressed, and those changes cannot be seen by just looking at the DNA sequence.

3. What is the difference between genomic DNA and the RNA molecules measured by RNA-seq?

Genomic DNA contains the full set of genetic information in every cell and stays the same throughout the plant's life and in all conditions. RNA, on the other hand, is the transcribed copy of

active genes. RNA-seq measures mRNA, which represents the genes that are currently being expressed, whereas DNA represents all genes whether active or not.

4. What is a biological replicate and why is it important?

A biological replicate is an independent sample in this study, it means leaves from a separate plant grown and treated under exactly the same conditions. Biological replicates are important because they account for natural variation between individual plants. They help confirm that observed differences are truly caused by salinity stress and not just by chance or variation in one particular plant. They also make the results statistically reliable.

5. What is the difference between single-end and paired-end sequencing?

In single-end sequencing, we read only from one end of the DNA fragment, producing one sequence per fragment. In paired-end sequencing, we read from both ends the forward (R1) and reverse (R2), producing two sequences that point toward each other. Paired-end reads provide more information, improve alignment accuracy, and help detect structural variations, which is why it was used in this study.

6. What is a FASTQ file?

A FASTQ file is the standard output format from high-throughput sequencing. It stores both the nucleotide sequence (the actual A, T, C, G bases) and a quality score for each base, indicating how confident the sequencer is that the base was called correctly. Each sequence entry has four lines: an identifier, the sequence itself, a separator line, and the quality scores encoded as characters.

7. What information does FastQC provide?

FastQC is a quality control tool that generates a comprehensive report to assess the quality of raw sequencing data. It provides statistics such as total number of reads, read length, GC content, per-base quality scores, sequence duplication levels, presence of adapters, and overrepresented sequences. It helps us check if the data is clean, reliable, and suitable for analysis.

8. What does a high per-base quality score indicate?

A high per-base quality score means the base call at that position is highly accurate and reliable. Quality scores are on a logarithmic scale: Q30 means a 1 in 1,000 chance of an incorrect base call. High scores across all positions indicate good sequencing quality and give confidence that the sequence data is trustworthy.

9. Why can adapter contamination be a problem?

Adapter sequences are short DNA fragments attached to the ends of the DNA during library preparation. If they are not completely removed before sequencing, they may appear in the reads. This is a problem because adapters are not part of the organism's genome, if left in the data, they interfere with read alignment, cause mismatches, and can lead to incorrect or unreliable results during downstream analysis.

10. Were all RNA-seq samples in your group similar in quality? Explain.

Yes, all samples showed very similar and excellent quality. Every sample had consistently high per-base quality scores (approximately Q35–Q36) across the entire read length, uniform read length of 101 bp, no adapter contamination, and similar GC content around 43–45%. Read counts were also comparable at approximately 28–31 million reads per sample. The only minor difference

was one R2 file showing a very low level of poly-G sequence, which is a harmless sequencing artifact and not a true quality issue.

11. Did any sample show a possible quality problem? What was it?

One R2 file showed a yellow warning flag for overrepresented sequences, a short poly-G sequence appeared at a very low level of only 0.24%. This is not a real problem, it is a well-known artifact specific to Illumina sequencing caused by signal decay in the G nucleotide channel, and it is not biological contamination. The percentage is extremely low and does not affect the reliability of the data.

12. What additional steps would be needed before the researchers could compare gene expression between control and treatment samples?

First, trim low-quality bases and remove adapters. Then, map clean reads to a reference or assemble a transcriptome. Next, count reads per gene, normalize for differences in sample size, and use statistical tests to find genes with significant expression changes between groups.

Part 14. GitHub Documentation

GitHub Group Repository Link:

https://github.com/ryanpas123e/Group4_RNASeq_Familiarization.git

Part 15. Group Summary

1. What study did the group choose and why is it relevant to the assigned topic?

Our group selected "Transcriptomic analysis reveals candidate genes associated with salinity stress tolerance during the early vegetative stage in faba bean genotype, Hassawi-2" by Afzal et al. (2023, *Scientific Reports*). This study directly matches our assigned topic of salinity stress, since it examines how a plant's gene expression shifts in response to salt exposure over time, comparing an unstressed control group to salt-treated samples at five different time points

2. What organism, tissue, control, and treatment were used?

The organism is faba bean (*Vicia faba* L.), specifically the salt-tolerant genotype "Hassawi-2." The tissue sampled was leaf tissue from seedlings at the early vegetative (three-leaf) stage. The control group consisted of untreated seedlings, while the treatment group was exposed to 200 mM NaCl and sampled at five time points after treatment: 6, 12, 24, 48, and 72 hours.

3. How many RNA-seq samples were included in the original study?

The study included 18 total samples with six conditions (control plus five stress time points), each with three biological replicates. On NCBI SRA, however, only 17 of these runs are currently listed; one replicate (Stress 48h, Rep 2) does not appear to be publicly available.

4. What sequencing platform and read type were used?

The authors used Illumina paired-end sequencing (HiSeq 4000), with a read length of 101 bp per read.

5. What did the group learn from examining the raw RNA-seq data?

Working directly with the FASTQ files and FastQC reports taught us how to interpret real sequencing quality metrics rather than just reading about them. We experienced firsthand that RNA-seq data has some "expected" quirks like biased base composition at the start of reads and high sequence duplication that would look alarming in genomic DNA sequencing but are normal and expected for RNA-seq, since highly expressed genes naturally produce many duplicate reads. We also learned how much manual effort goes into consolidating data across a group when everyone works in separate accounts, since Galaxy doesn't support real-time shared histories the way we initially assumed.

6. Were there noticeable differences in data quality among samples?

Overall, all five group members' samples were very consistent in quality, similar read counts (roughly 28–32 million reads per file), similar GC content (42–46%), and no adapter contamination in any sample. One sample, Batigulao's (SRR23869383, 24h stress), stood out slightly with a marginally lower FastQC fail rate and the highest GC content and duplication rate in the group, but this was a minor difference rather than a quality concern.

7. What analysis did the original authors perform after quality control?

After quality filtering and trimming (using Trimmomatic), the authors performed de novo transcript assembly with Trinity, since no reference genome was used. They then clustered and annotated the resulting transcripts using tools like CD-HIT-EST, TransDecoder, and BLAST-based searches against databases such as NR, Swiss-Prot, KEGG, and GO. Differential expression between control and each stress time point was calculated using edgeR, and selected results were validated experimentally with qRT-PCR.

8. What will the group need to learn next in order to reproduce part of the authors' gene expression analysis?

Since our group only has computers and Galaxy, all next steps stay fully computational, just like this assignment. We'll need to learn how to trim reads (e.g., with Trimmomatic), then either align to a reference genome or perform de novo assembly with Trinity, as the original authors did. From there, we'll need to generate expression counts and use a tool like edgeR to statistically compare control and treatment samples. All of this happens through Galaxy's web-based tools, the same way we ran FastQC here where no wet-lab work required.

References

Afzal, M., Alghamdi, S. S., Khan, M. A., Al-Faifi, S. A., & Rahman, M. H. U. (2023). Transcriptomic analysis reveals candidate genes associated with salinity stress tolerance during the early vegetative stage in fababean genotype, Hassawi-2. *Scientific Reports*, 13(1), 21223. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692206/>

Stress 24h-First paired-end file R1 - SRA - NCBI. (n.d.). <https://www.ncbi.nlm.nih.gov/sra/SRX19681430%5Baccn%5D>